A balanced journal can still be $1 short.

Double-entry accounting checks internal consistency.
It cannot prove every expected payment arrived.

THE QUESTION Did the processor settle the full amount?

First delivery

6 inserted



Same batch replayed

0 new rows

Same source + same batch ID

→ 6 duplicates detected → no second financial effect

A reliable import is safe to retry.

Idempotency protects against duplicate delivery; it does not reconcile settlement.

PAYMENT RECONCILIATION / 03

Compare expected with settled

P3 / USD cents

Expected	**2,500 ($25.00)**
Processor	**2,400 ($24.00)**
Difference	**−100 cents (−$1.00)**

P9 / unmatched

External amount

$5.00

No internal payment found

Keep two exceptions distinct:

P3 is short by $1.00. P9 has no expected amount to compare.

PAYMENT RECONCILIATION / 04

Three controls answer different questions

1 REPLAY

Did we load this batch before?

Output: Duplicates

2 JOURNAL

Do debits equal credits?

Output: Balanced

3 SETTLEMENT

Did the processor pay what we expected?

Output: Exceptions

A green result in one layer does not clear the next layer.

TRY THE EXERCISE / 05

Run the reconciliation locally

Python 3.12+ · Git · no paid cloud account or API key

```
python -m fintech_lab.quickstart
```

The example data is synthetic and simplified for learning.

100-lesson fintech data engineering roadmap

github.com/antony502189-max/fintech-data-engineering-roadmap

Use it freely. If it helps, a voluntary star is appreciated.